

Education

Inria Paris (ALMAnaCH team) | Sorbonne University

Paris, France

PHD CANDIDATE

Mar. 2023 - Mar. 2026

- Research Topic: Patent Representation Learning for Innovation Generation and Technical Trend Analysis.
- Supervisors: Benoît Sagot, Eric Villemonte de la Clergerie and Kim Gerdes.

University Paris-Saclay

Orsay, France

2ND YEAR OF MASTER IN ARTIFICIAL INTELLIGENCE

Sep. 2020 - Sep. 2021

- GPA: 16.7/20 (Rank: 1/17)
- Main courses: Generative Models, Graphical Models, Natural Language Processing, Multilingual NLP, Advanced Optimisation, etc.

ENSIIE (École Nationale Supérieure d'Informatique pour l'Industrie et l'Entreprise)

Évry, France

ENGINEER DIPLOMA

Sep. 2018 - May 2021

- Specialized engineering school of Mines Télécom group.
- Majored in Applied Mathematics and Computer Science.

Xidian University

Shaanxi, China

BACHELOR DEGREE - INFORMATION AND COMPUTING SCIENCE

Aug. 2015 - July 2018

- GPA: 3.60/4.0
- Main courses: Mathematical analysis, Matrix theory, Optimisation, Functional analysis, Probability theory, Statistics, etc.

Professional Experience

QATENT (A Questel company)

Paris, France

R & D ENGINEER — NLP FOR PATENT ANALYSIS

Dec. 2022 - Present

- Trained and implemented Language Model Models (LLMs) to enhance patent generation and established a Retrieval-Augmented Generation (RAG) system for patent prior-art search.
- Conducted Open Information Extraction (OpenIE) on patent texts and spearheaded the creation of a knowledge graph.
- Designed and developed domain-specific multilingual taxonomies to optimize information organization and retrieval.

Inria Paris - ALMAnaCH

Paris, France

RESEARCH ENGINEER — LARGE-SCALE MULTI-LABEL PATENT CLASSIFICATION

Oct. 2021 - Nov. 2022

- Collaborated with the National Institute of Industrial Property (INPI) of France to develop an automated patent classification system with a label space of 6k and 70k categories.
- Conducted comparative analysis and optimization of classification methods on both English and French corpora, including one-vs-all classifiers using sparse features (TF-IDF, BOW), pre-trained language models (Roberta, XLNet, Bert-for-patents), and Extreme Multi-label Learning (Parabel, AttentionXML, LightXML, etc.).

CNRS - LISN

Orsay, France

RESEARCH INTERN — TEXT-MINING AND ONTOLOGY CONSTRUCTION ON PATENT DATA

Mar. 2021 - Sep. 2021

- Developed a technological terminology recognition system with high accuracy and generality for classes of patents (CPC domains A to H), including solutions for overlapping terms, by training a spaCy named entity recognition (NER) model based on a dataset constructed by a rule-based matcher.
- Extracted and managed term definitions from dumps of English Wiktionary and Wikipedia using an SQLite database.
- Implemented and trained state-of-the-art models (TransE, M3GM, and T5) for predicting direct hypernym/hyponym relationships, as well as different seq2seq models to generate hypernym/hyponym paths.

JD.COM

Beijing, China

DATA ANALYSIS INTERN

June 2020 - Aug. 2020

- Extracted and computed data indicators from Hadoop, developed ETL tables, and transferred results to the BI platform. Created data visualizations to effectively communicate insights to stakeholders.
- Collaborated with product managers and other departments to implement AB testing and iteratively improve product services. Used data storytelling to communicate results and recommendations to non-technical stakeholders.

Skills

Programming

Python, R, SQL, Shell, OCaml, C++

DL / NLP / LLM frameworks

Pytorch, TensorFlow, Transformers, spaCy, Gensim, NLTK, peft, vLLM, LangChain

Languages

Chinese (Native), French (C1), English (C1)

CONFERENCE PROCEEDINGS

PatentEval: Understanding Errors in Patent Generation

You Zuo, Kim Gerdes, Éric Clergerie, Benoît Sagot

Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), 2024

Exploring Data-Centric Strategies for French Patent Classification: A Baseline and Comparisons

You Zuo, Benoît Sagot, Kim Gerdes, Houda Mouzoun, Samir Ghamri Doudane

18e Conférence en Recherche d'Information et Applications \ 16e Rencontres Jeunes Chercheurs en RI \ 30e Conférence sur le Traitement Automatique des Langues Naturelles \ 25e Rencontre des Étudiants Chercheurs en Informatique pour le Traitement Automatique des Langues, 2023

Technological taxonomies for hypernym and hyponym retrieval in patent texts

You Zuo, Yixuan Li, Alma Parias García, Kim Gerdes

ToTh 2022 - Terminology & Ontology: Theories and applications, 2022, Chambéry, France

Patent Classification using Extreme Multi-label Learning: A Case Study of French Patents

You Zuo, Houda Mouzoun, Samir Ghamri Doudane, Kim Gerdes, Benoît Sagot

SIGIR 2022-PatentSemTech workshop-3rd Workshop on Patent Text Mining and Semantic Technologies, 2022